PRODUCT SPECIFICATION

MIRRORWORLD

Platform Upgrade Specification

Critical alerts, interactive digital campus, multi-variable simulation, MIRROR AI copilot and the Digital Twin Difference Engine

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2	Feature 01: MIRROR AI Decision Copilot	Natural-language interface to the engines
3	Digital Twin Difference Engine	Dependency-aware change analysis

Guiding principle. Every number shown to an administrator must be computed from the campus model, the simulation engine or the optimizer. Nothing is estimated by language models, and estimated values are always labelled as estimates.

PART 1 - UPGRADES TO EXISTING MODULES

1. Overview

The Overview becomes the place where administrators understand the campus's operational condition **before** entering the Simulation Studio.

A. Critical Alerts

The system automatically identifies important campus problems. Example conditions:

- Three classrooms are exceeding their assigned capacity.
- Two teachers have overlapping sessions.
- One laboratory has insufficient equipment availability.
- A particular building has unusually high modeled energy consumption.
- Several sessions are affected by a facility disruption.

Each alert must include:

#	Field	Purpose
1	Problem description	What is wrong, stated plainly
2	Severity level	Low / Medium / High / Critical
3	Affected resources	Rooms, labs, instructors, student groups, buildings
4	Potential consequences	What happens if the problem is left unresolved
5	Suggested next action	A concrete step the administrator can take

Example alert

Classroom overcapacity - HIGH. Three rooms in Building A are scheduled above capacity in the Tuesday 10:00 slot. **Affected:** A-101, A-204, A-305 and 4 student groups. **Consequence:** sessions may be moved or cancelled at short notice. **Next action:** review the proposed reassignment to larger rooms in Building B.

Illustrative only; actual alerts are generated from live campus data.

B. Decision Recommendations

Add a section called **Recommended Actions**.

> *"Building A has significantly higher room utilization than Building B. Consider redistributing selected sessions to Building B."*

Requirement. Recommendations must be based on actual campus data and calculated metrics (for example utilization ratios, capacity margins, conflict counts and modeled energy use), never on generic advice.

PART 1 - UPGRADES TO EXISTING MODULES

2. Digital Campus

D. Student Movement Visualization

Represent movement between buildings during timetable transitions. For example, students finish a class in Building A and need to attend another class in Building C. The system estimates:

Metric	Description
Number of students moving	Derived from enrollments of consecutive sessions
Distance between buildings	Calculated from building locations on the campus model
Expected movement time	Distance divided by an assumed walking speed, compared with the gap between sessions
Potential congestion	Flow volume along shared paths during the transition window

Visualization approach

- Animated paths on the campus map, with thickness or intensity showing the number of students.
- A time control to step through timetable transitions across the day.
- Highlighting of tight transitions where the travel time exceeds the gap between sessions.

Labelling rule. Start with estimated movement based on the timetable and building locations. Do **not** label it real-time tracking unless live movement data actually exists. Use wording such as "Estimated movement" or "Timetable-based flow".

Recommendation: make Digital Campus visually interactive, not just a static map.

PART 1 - UPGRADES TO EXISTING MODULES

3. Simulation Studio

A. Disruption Categories

Each category dynamically displays its relevant configuration fields.

Category	Scenario
Instructor	Teacher unavailable, or workload changed
Infrastructure	Building or equipment unavailable
Emergency	Multiple operational disruptions occurring together

B. Multi-Variable Simulation

Extremely important. The form currently focuses on individual disruption parameters. Users must be able to combine multiple changes into one scenario, because operational decisions rarely affect only one variable.

Scenario: Semester Expansion

- Increase student enrollment by 200.
- Make two classrooms unavailable.
- Reduce laboratory availability by 25%.
- Extend campus operating hours by one hour.

The system evaluates all four changes **together**, not one at a time.

C. Scenario Templates

Ready-made simulation templates. Selecting one loads the relevant parameters automatically, and the user can then customize them.

#	Template	Typical parameters loaded
1	Enrollment Expansion	Student increase, program, intake term
2	Laboratory Shutdown	Lab selection, duration, equipment loss
3	Teacher Shortage	Instructors unavailable, workload limits
4	Classroom Overcrowding	Room capacity thresholds, group sizes
5	Energy Conservation	Building schedules, energy targets, operating hours
6	Emergency Campus Disruption	Combined building, staff and equipment outages
7	Semester Timetable Restructuring	Slot structure, session lengths, operating hours

D. Impact Severity Prediction

After the disruption parameters are entered, show a preliminary impact assessment calculated from the dependency relationships and the current campus model.

Potential Impact: HIGH

- 12 sessions may be affected.
- 3 classrooms may become unavailable.
- 2 instructors may require reassignment.
- 4 student groups may experience timetable changes.

Illustrative figures. Real values come from the dependency model.

PART 2 - COMPLETELY NEW FEATURES

Feature 01: MIRROR AI

Intelligent Decision Copilot

An AI-powered interface through which administrators communicate with the digital twin in natural language. Instead of manually configuring every simulation, users describe their problem.

> *"Next semester, we expect 200 additional CSE students. We cannot construct new classrooms. Find a feasible timetable that minimizes scheduling conflicts and estimated energy consumption."*

What the AI does

1. Interpret the user's request.
2. Extract the required parameters.
3. Identify missing information.
4. Create a proposed simulation configuration.
5. Send it to the actual simulation engine.
6. Run the optimization process.
7. Retrieve the calculated results.
8. Explain the findings in natural language.

Additional capabilities

- Ask questions about campus conditions.
- Explain why a timetable is infeasible.
- Summarize simulation results.
- Identify frequently occurring resource problems.
- Explain differences between scenarios.
- Generate decision reports.

Important technical requirement. The AI must never invent results. All numerical results must come from the actual simulation and optimization engines. MIRROR AI is an interface to MIRRORWORLD's computational systems, not an independent chatbot.

Request flow

Stage	Component	Responsibility
1	Language layer	Interprets intent, extracts parameters, asks for missing inputs
2	Configuration builder	Produces a structured simulation configuration for user confirmation
3	Simulation and optimization engines	Compute all metrics, schedules and conflicts
4	Language layer	Explains the returned results, quoting only engine outputs

PART 3 - CORE CAPABILITY

Digital Twin Difference Engine

A system that identifies how the proposed campus differs from its original state and explains the operational consequences of every meaningful difference. It is intended as a **core capability** rather than an isolated page.

Worked example

Attribute	Original state	New state
Classrooms	20	20
Laboratories	4	3 available
Students	1,000	1,200
Scheduled sessions	112	112 (before optimization)

Change	Resulting implication
Student population increased	More capacity may be required
Laboratory availability decreased	Some sessions may require reassignment
Classroom count unchanged	Existing infrastructure must accommodate additional demand
Session requirements unchanged	Scheduling constraints become tighter

The Difference Engine explains

1. What changed.
2. Why it matters.
3. Which resources are affected.
4. Which constraints become more difficult.
5. What decisions could address the problem.

It then generates a **dependency-aware impact report**.

System integration

The Difference Engine connects four existing components into one coherent system. It is shared infrastructure: Overview alerts, Simulation Studio severity prediction and MIRROR AI explanations should all draw their change analysis from it.

Component	Role in the Difference Engine
Campus Model	Source of the original and proposed states
Dependency Explorer	Traces which resources and sessions each change touches
Simulation Studio	Evaluates scenarios that produce the proposed state
Schedule Optimizer	Tests whether a feasible timetable exists and at what cost